

Implementing a Compiler

Kenjiro Taura

Contents

The MinC (“Minimum C”) language	2
Overview of Inside a Compiler	4
Lexer and parser : source code $\rightarrow$ AST	10
Code generation	38
Intermediate Representation (IR)	56

The MinC (“Minimum C”) language

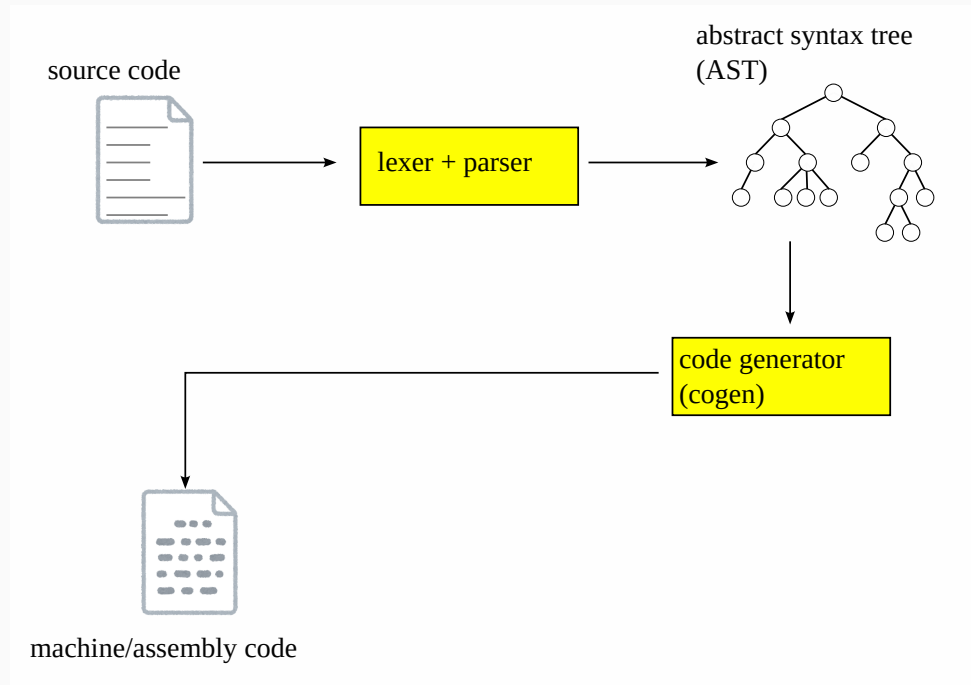
MinC (“Minimum C”) spec overview

- all expressions have type `long` (64 bit integer)
 - no other integers, floating point numbers, pointers, or structs
 - everything is long $\Rightarrow$ *type checks are unnecessary*
- no global variables or `typedef`
 - $\Rightarrow$ a program = list of *function definitions*
- supported complex statements are `if`, `while`, and compound statement (`{ ... }`) only
- function calls follow the C convention $\Rightarrow$ MinC code can call or be called by functions compiled by other compilers (e.g., gcc)

Overview of Inside a Compiler

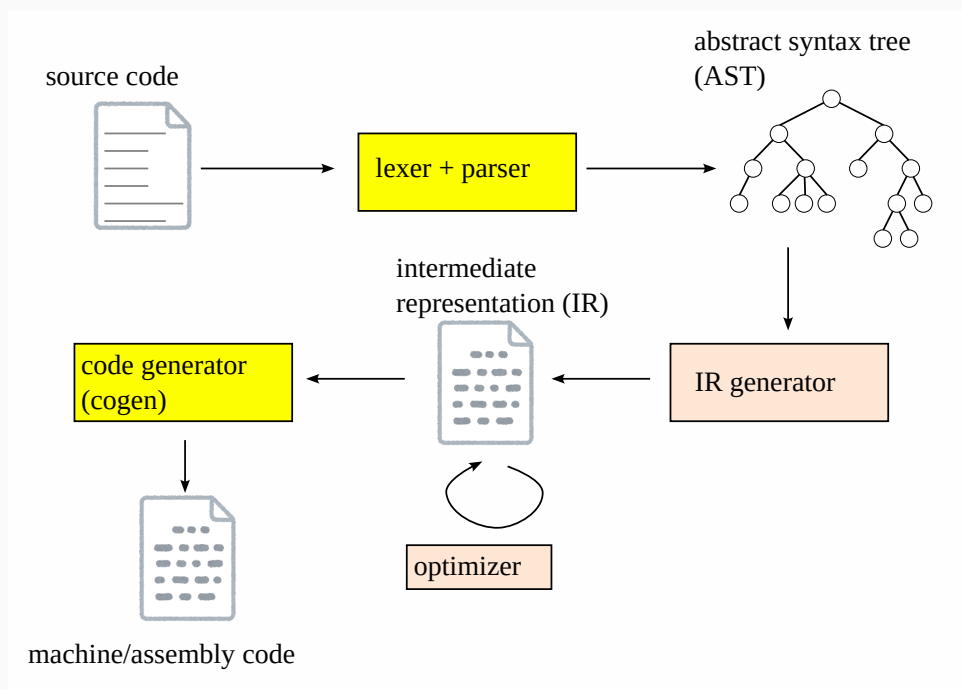
Data structures

- **Abstract Syntax Tree (AST):** data structure representing the program



Data structures

- **Abstract Syntax Tree (AST):** data structure representing the program
- **Intermediate Representation (IR):** common representation portable across multiple source/target languages



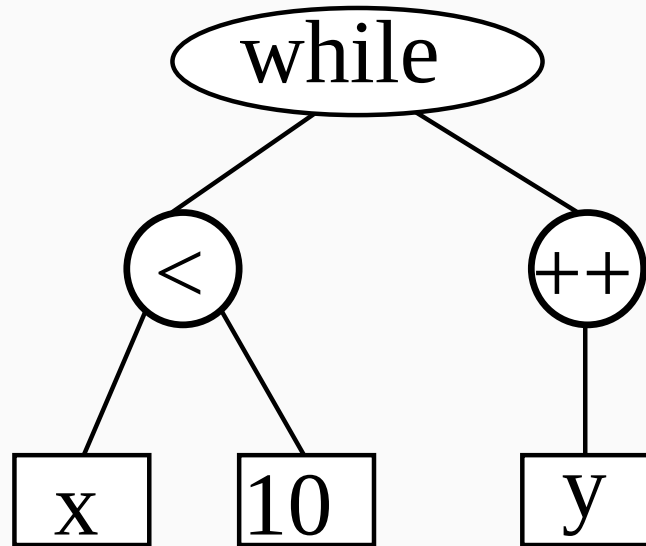
Typical compilation steps

1. **lexing and parsing:** source code (string) $\rightarrow$ AST
2. IR generation: AST $\rightarrow$ IR (*)
3. optimization: IR $\rightarrow$ IR (*)
4. **code generation:** IR $\rightarrow$ assembly

(*) : optional steps

Abstract Syntax Tree (AST)

- a data structure that naturally represents a program
- expression,
- statement,
- function definition,
- the whole program,
- ...



- also called **parse tree**

Components of the baseline code

- `{go,jl,ml,rs}/minc/`

<code>lex.??</code>	lexer or tokenizer (string $\rightarrow$ stream of tokens)
<code>ast.??</code>	abstract syntax tree (AST) definition
<code>parse.??</code>	parser (stream of tokens $\rightarrow$ AST)
<code>codegen.??</code>	AST $\rightarrow$ assembly
<code>main.??</code> or <code>minc.??</code>	main driver

Your work

1. read given, understand, and explain how they work
2. {lex, ast, parse} all lack code for `while` statement
 - $\Rightarrow$ add necessary definitions for `while` statement
3. implement `codegen`, which is almost empty

**Lexer and parser : source
code $\rightarrow$ AST**

Lexer and parser

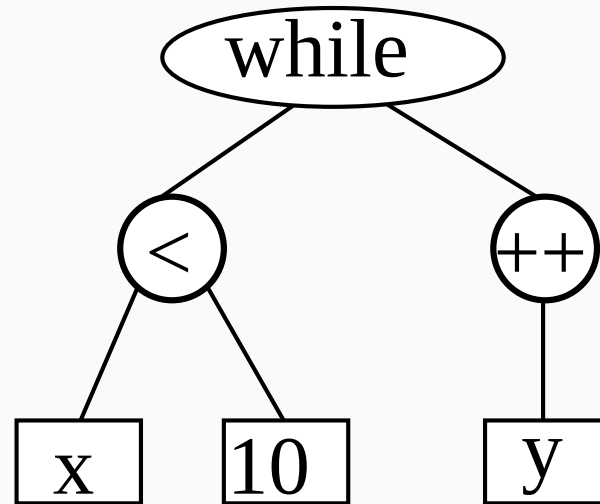
- **lexer**: string $\rightarrow$ sequence of *tokens* ($\approx$ words)
 - also called *lexical analyzer*, or *tokenizer*
 - `while (x < 10) y++;  $\Rightarrow$`

while	(	x	<	10	)	y	++	;
WHILE	LP	ID	CMP	INT	RP	ID	PLSPLS	SEMICOLON

Lexer and parser

- **parser:** sequence of tokens $\rightarrow$ AST

while	(	x	<	10	)	y	++	;
WHILE	LP	ID	CMP	INT	RP	ID	PLSPLS	SEMICOLON

 $\Rightarrow$ 

Specifying a grammar

- a grammar for *tokens*
 - specifies which character sequence constitutes a valid token
 - typically uses *Regular Expressions (RE)*
- a grammar for *the entire program*
 - specifies which token sequence constitutes a valid input
 - typically uses (a subset of) *Context Free Grammar (CFG)*
- note: there is an approach that uses a single grammar for both

Regular expression

- a regular expression is any expression that can be formed by:

ε	(empty string)
c	(a character)
$E E$	(concatenation)
$E \mid E$	(alternation)
E^*	(zero or more repetition)
(E)	(paren)

where E is a regular expression

- $\mid$, * , $($ and $)$ are literals

Regular expression

- expressions for convenience

$E^+ \equiv E E^*$ (one or more repetition)

$E? \equiv \varepsilon \mid E$ (optional)

Regular expression examples

- to build complex expressions, use symbols to represent regular expressions used in other regular expressions. e.g.,

nz	=	1 2 3 4 5 6 7 8 9	1, 2, ..., 9
digit	=	0 nz	0, 1, 2, ..., 9
non_neg	=	0 nz digit*	0, 12, 34
int	=	-? non_neg	0, -0, 12, -34
fraction	=	int (. digit*)?	-12.34
float	=	fraction (e int)	-12.34e-5
alpha	=	A B ... Z a b ... z	A, B, ..., Z, a, b, ..., z
alpha_	=	alpha _	A, B, ..., Z, a, b, ..., z, _
id	=	alpha_ (alpha_ digit)*	a, abc, a0_b1

Regular expression semantics (just for formality ...)

- a regular expression E represents *a set of strings*, written $\llbracket E \rrbracket$

$$\llbracket \epsilon \rrbracket = \{ \text{“”} \}$$

$$\llbracket c \rrbracket = \{ c \}$$

$$\llbracket E_0 E_1 \rrbracket = \{ e_0 + e_1 \mid e_0 \in \llbracket E_0 \rrbracket, e_1 \in \llbracket E_1 \rrbracket \}$$

$$\llbracket E_0 \mid E_1 \rrbracket = \llbracket E_0 \rrbracket \cup \llbracket E_1 \rrbracket$$

$$\llbracket E^* \rrbracket = \{ \text{“”} \} \cup \{ e_0 + e_1 \mid e_0 \in \llbracket E \rrbracket, e_1 \in \llbracket E^* \rrbracket \}$$

$$\llbracket (E) \rrbracket = \llbracket E \rrbracket$$

- note: “+” represents string concatenation

Context Free Grammar (CFG)

- specified by a collection of *production rules*
- a production rule looks like

$$L \rightarrow R_0 R_1 \dots$$

where

- L : a symbol (*non-terminal*)
- R_i is either
 - a symbol defined by a production rule(s), or
 - a token name (a *terminal* symbol)

An example : expressions

$\text{expr} \rightarrow$	int	12, 345, ...
$\text{expr} \rightarrow$	id	f, x, i, is_prime, ...
$\text{expr} \rightarrow$	unop expr	-x, exp, !a_greater_than_b
$\text{expr} \rightarrow$	expr binop expr	x + y, a * x + b * y + 1, a & b, ...
$\text{expr} \rightarrow$	(expr)	3 * (a + 1)
$\text{expr} \rightarrow$	funcall	

- blue symbols (int, id, unop, binop, (,)) are terminals (tokens)
- above rules overlook the fact that some operators (i.e., + and -) can be used as a unary operator and a binary operator

An example : function call

funcall	→	id (comma_exprs)	f(x, 2 * y, 1)
comma_exprs	→		
comma_exprs	→	expr	
comma_exprs	→	expr comma_expr_star	
comma_expr_star	→		
comma_expr_star	→	, expr comma_expr_star	

An example : statements

stmt $\rightarrow$;
stmt $\rightarrow$ continue ;
stmt $\rightarrow$ break ;
stmt $\rightarrow$ return ;
stmt $\rightarrow$ { decl* stmt* }
stmt $\rightarrow$ if (expr) stmt (else stmt)?
stmt $\rightarrow$ while (expr) stmt
stmt $\rightarrow$ expr ;

- as you have seen,
 - the same symbol L can appear multiple times in the lefthand side (i.e., *alternation*)
 - R_i can be L or any symbol defined earlier or later (i.e., definitions can be *recursive*)

A few shorthands

- we often use shorthands (`|`, `?`, `*`, `+`) that have similar meanings with those for RE
- they can be mechanically eliminated
- the above example using the shorthands:

```
expr      → int | id | unop expr | expr binop expr | funcall
funcall   → id ( comma_exprs )
comma_exprs → | expr ( , expr )*
```

CFG semantics (for formality)

- each symbol L represents a set of token sequences ($\llbracket L \rrbracket$)
- $\llbracket L \rrbracket$ is the set of token sequences that can result by, starting from L , repeatedly replacing a non-terminal symbol to the righthand side of its production rule, until it becomes a sequence of tokens (terminals)

```
expr  → funcall
      → id ( comma_exprs )
      → id ( expr comma_expr_star )
      → id ( id comma_expr_star )
      → id ( id , expr comma_expr_star )
      → id ( id , expr + expr comma_expr_star )
      → id ( id , id + expr comma_expr_star )
```

CFG semantics (for formality)

→ $\text{id (id , id + int comma_expr_star)}$

→ $\text{id (id , id + int)}$

∴ $\text{id (id , id + int)} \in \llbracket \text{expr} \rrbracket$

e.g., $\text{f (x, y + 1)} \in \llbracket \text{expr} \rrbracket$

An alternative semantics

- $\llbracket . \rrbracket$ is the minimal set of token sequences satisfying:

1. $\llbracket t \rrbracket = \{ t \}$ (t : terminal)

2. $L \rightarrow R_0 \dots R_{n-1}$ implies

$$\begin{aligned} r_0 \in \llbracket R_0 \rrbracket, \dots, r_{n-1} \in \llbracket R_{n-1} \rrbracket \\ \Rightarrow r_0 + \dots + r_{n-1} \in \llbracket L \rrbracket \end{aligned}$$

- “+” represents concatenation of token sequences

CFG is more expressive than RE

- as you might have noticed, RE is a special case of CFG
- all the constructs of RE can be straightforwardly expressed with CFG, where a token = a character
- e.g., a CFG equivalent to RE “int = 0 | nz digit^{*}”

int $\rightarrow$ 0

digit $\rightarrow$ 0

int $\rightarrow$ nz digits

digit $\rightarrow$ nz

digits $\rightarrow$

nz $\rightarrow$ 1 | ... | 9

digits $\rightarrow$ digit digits

In general ...

- below, $\text{CFG}(e, L)$ is a function that converts regular expression e to an equivalent CFG s.t., $\llbracket L \rrbracket = \llbracket e \rrbracket$

$$\text{CFG}(\varepsilon, L) = \{L \rightarrow\}$$

$$\text{CFG}(c, L) = \{L \rightarrow c\}$$

$$\text{CFG}(E_0 E_1, L) = \{L \rightarrow R_0 R_1\} \cup \text{CFG}(E_0, R_0) \cup \text{CFG}(E_1, R_1)$$

$$\text{CFG}(E_0 \mid E_1, L) = \{L \rightarrow R_0, L \rightarrow R_1\} \cup \text{CFG}(E_0, R_0) \cup \text{CFG}(E_1, R_1)$$

$$\text{CFG}(E^*, L) = \{L \rightarrow \mid R L\} \cup \text{CFG}(E, R)$$

$$\text{CFG}((E), L) = \text{CFG}(E, L)$$

- R, R_0 and R_1 are unique symbols that do not appear elsewhere

A CFG that cannot be expressed by RE

- intuitively, RE *can repeat* (E^*) *but cannot recurse*
- e.g., both “ $A \rightarrow \mid a A$ ” and “ $A \rightarrow \mid A a$ ” *can* be expressed by an RE (both are equivalent to a^*), but

$$A \rightarrow \mid a A b$$

cannot ($\llbracket A \rrbracket = \{\varepsilon, ab, aabb, aaabbb, \dots\} = \{a^n b^n \mid n \geq 0\}$)

- **the proof** is interesting but omitted

If $RE \subset CFG$, why use both (not just CFG)?

- parsing *general* CFG is expensive ($O(\text{length}^3)$)
- the primary reason is handling *alternatives* may require *backtrack*

$$A \rightarrow B_0 B_1 \dots \mid C_0 C_1 \dots \mid D_0 D_1 \dots$$

- practical parsers take either of the following two approaches
 1. allow only alternatives that can be determined with a *limited lookahead* (*LL(1), LALR(1), etc.*)
 2. allow backtrack with programmer-supplied *cut points* (*Parsing Expression Grammar; PEG*)

CFG with a limited lookahead (LL(1), LALR(1), etc.)

- recall the syntax of statement

stmt $\rightarrow$; | continue ; | break ; | return ; | { decl* stmt* }
| if (expr) stmt (else stmt)? | while (expr) stmt
| expr ;

- upon parsing a statement, which branch we should take can be determined just by its *first* token
- it is essential to have a separate tokenizer for this type of grammar*
(looking ahead a token $\neq$ looking ahead a character)

How to write a CFG parser in practice

1. use parser-generator tool

- yacc/bison (C/C++)
- Menhir (OCaml)
- ANTLR4
- JavaCC
- ...

2. *manually write LL(1) parser*

- This course chooses the latter for lack of a parser generator common across the four languages
- The experience helps you write a robust parser in other occasions

How to write LL(1) parser?

- For each symbol A (terminal or non-terminal), define a function,

`parse_A(tokens)`,

which takes a stream of tokens and consumes a prefix of it constituting A

- **`tokens`** supports the following interface
 - **`peek(tokens)`** : the “current” (the first unconsumed) token
 - **`advance(tokens)`** : consume the current token

How to write LL(1) parser?

1. If A is a terminal (a token) $\Rightarrow$ just consume it or raise an error

```
parse_A(tokens) {  
    if (peek(tokens) != A) parse_error();  
    advance(tokens);  
}
```

2. If A is a non-terminal and its definition is $A \rightarrow B C D$ (without alternatives), call corresponding **parse_??** functions in turn

```
parse_A(tokens) {  
    parse_B(tokens);  
    parse_C(tokens);  
    parse_D(tokens);  
}
```

Example: if statement

- $\text{if_stmt} \rightarrow \text{IF LPAREN expr RPAREN stmt (ELSE stmt)?}$

```
parse_if(tokens) {  
    parse_IF(tokens);      // consume IF token  
    parse_LPAREN(tokens);  // consume '(' token  
    cond = parse_expr(tokens); // consume expression  
    parse_RPAREN(tokens);  // consume ')' token  
    then_s = parse_stmt(tokens);  
    if (peek(tokens) == ELSE) {  
        parse_ELSE(tokens);  
        else_s = parse_stmt(tokens);  
    } else {  
        else_s = none  
    }  
    return if_stmt(cond, then_s, else_s)  
}
```

Handle alternatives

3. If A has alternatives, e.g.,

- $A \rightarrow B C$
- $A \rightarrow D E$

we choose the appropriate branch by

“peeking the current token”

```
parse_A(tokens) {  
  if (peek(tokens) in FIRST(B C)) {  
    parse_B(tokens);  
    parse_C(tokens);  
  } else if (peek(tokens) in FIRST(D E)) {  
    parse_D(tokens);  
    parse_E(tokens);  
  } else { parser_error(); } }
```

- $\text{FIRST}(\cdot)$ is the set of tokens that could start ‘ $\cdot$ ’
- The code assumes $\text{FIRST}(B C)$ and $\text{FIRST}(D E)$ do not overlap (LL(1))
- You can compute $\text{FIRST}(\cdot)$ by an algorithm, but for simple cases manual inspection suffices

Limitation of LL(1)

- It has trouble with “left recursion”
- e.g.,
 - $A \rightarrow$
 - $A \rightarrow A b$
- writing `parse_A` following the above formula would result in infinite recursion
- in practice, we get rid of left recursion by manual inspection
- e.g., the above is equivalent to b^* , which is
 - $A \rightarrow$
 - $A \rightarrow b A$

Code generation

Code generation (cogen) — basic structure

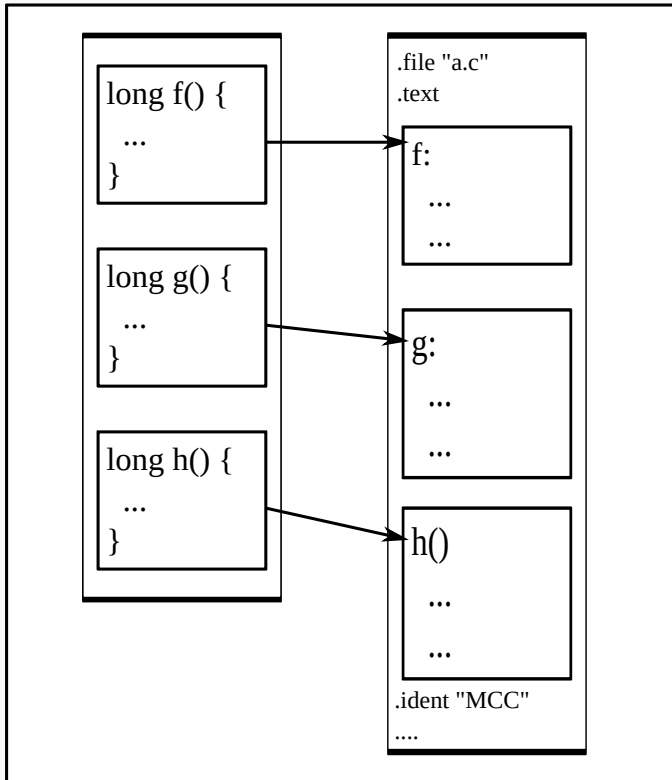
- takes an AST and returns machine code (a list of instructions)
- generate machine code for an AST $\approx$ generate machine code of its components and properly arrange them
- program $\rightarrow$ function definition $\rightarrow$ statement $\rightarrow$ expression

Code generation (cogen) — basic structure

- code generator has lots of:
 - case analysis based on the type of the tree; use:
 - pattern matching (match à la OCaml and Rust), or
 - polymorphism (OCaml objects, Julia function, Go interface, Rust trait)
 - recursive calls to child trees

Compiling an entire file

- $\approx$ concatenate compilation of individual function definitions



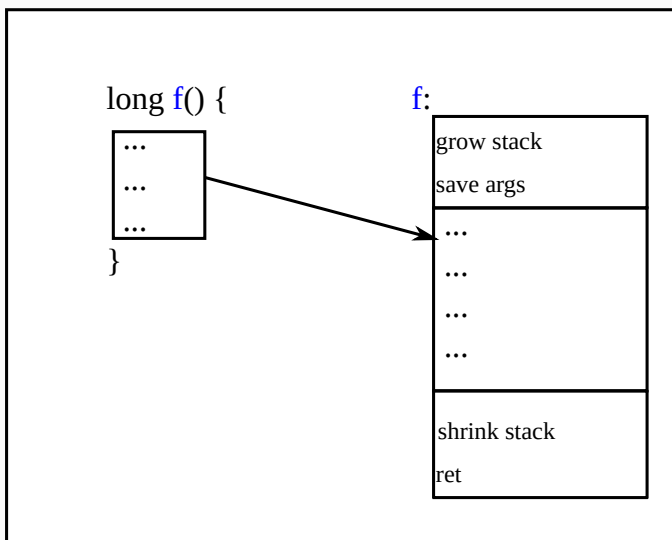
Pseudo code:

```
gen_program (Program([d0, d1, ...])) ... =  
    ...  
    header  
    + (gen_def d0 ...)  
    + (gen_def d1 ...)  
    + ...  
    + trailer
```

note: + is a list concatenation

Compiling a function definition

- $\approx$ prologue (grow the stack, etc.) + code for the body (statement) + epilogue (shrink the stack, `ret`, etc.)

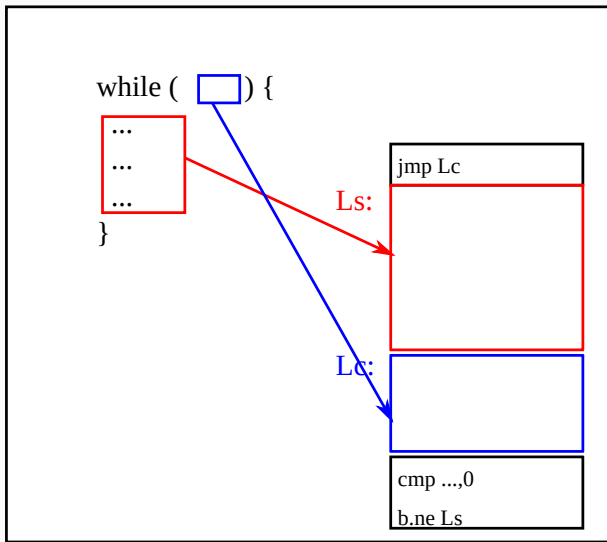


Pseudo code:

```
gen_def (DefFun(f, params, ret_type, body)) =  
  (gen_prologue f ...)  
  + (gen_stmt body ...)  
  + (gen_epilogue f ...)
```

Compiling a statement (while statement)

- $\approx$ jump to the condition expression; body; the condition expression; compare and conditional branch



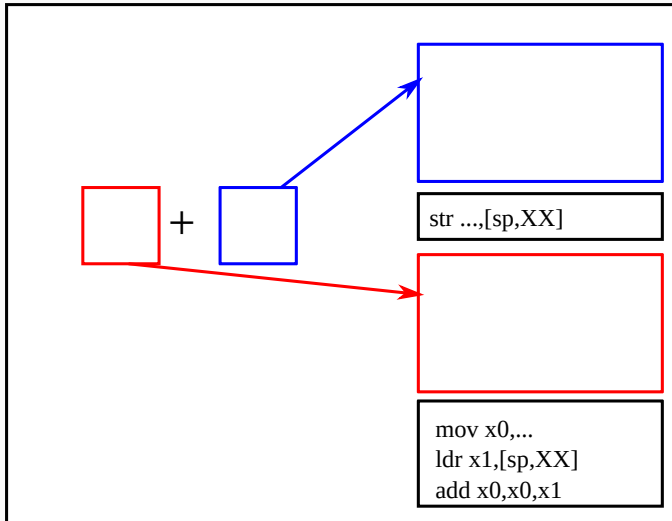
```
gen_while_stmt (StmtWhile(cond, body)) ... =  
    cond_op, cond_insns = gen_expr cond ... ;  
    body_insns = gen_stmt body ... ;  
    ...  
    [ jmp Lc; Ls ]  
+ body_insns  
+ [ Lc ]  
+ cond_insns  
+ [ cmp cond_op, 0; jne Ls ]
```

- (gen_expr *expr* ...) returns a pair: (instructions to evaluate *expr*, the location of the result)

Compiling an expression (arithmetic)

- $\approx$ instructions to evaluate the arguments + an appropriate arithmetic instruction
- “gen_expr E ...” returns a pair:

(instructions evaluation E , the location holding the value of E)



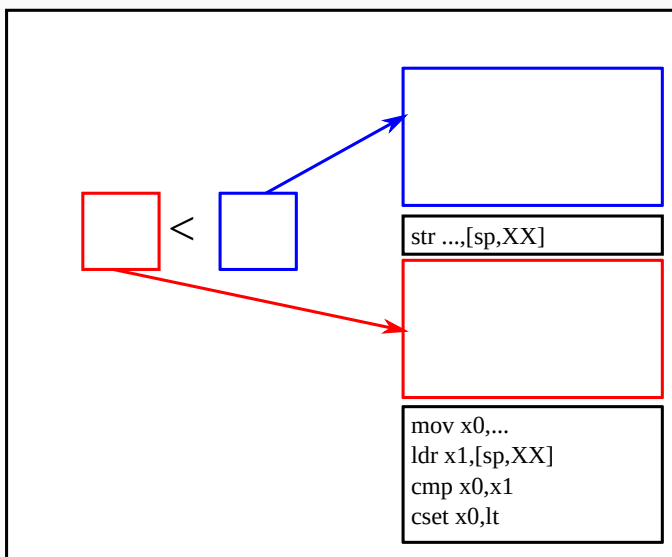
```
gen_add_expr ExprOp("+", [e0; e1]) ... =  
  insns1,op1 = gen_expr e1 ... ;  
  insns0,op0 = gen_expr e0 ... ;  
  m = (* a slot on the stack for e1 *);  
  ( insns1  
    + [ str op1,m ]  
    + insns0  
    + [ mov x0,op0;  
        ldr x1,m;  
        add x0,x0,x1 ],  
    x0)
```

Compiling an expression (comparison)

- $A < B$ is an expression that evaluates to:
 - 1 if $A < B$
 - 0 if $A \geq B$
- this can be done by `cmp` + conditional set (`cset`)

Compiling an expression (comparison)

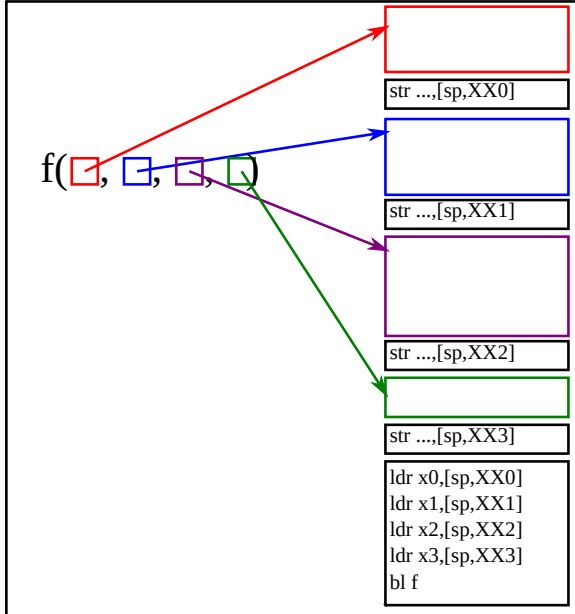
- $\approx$ compile the arguments; compare; conditional set



```
gen_cmp_expr (ExprOp("<", [e0; e1])) ... =  
  insns1,op1 = gen_expr e1 ... ;  
  insns0,op0 = gen_expr e0 ... ;  
  m1 = (* a slot on the stack for e1 *);  
  ...  
  (insns1  
    + [ str op1,m1 ]  
    + insns0  
    + [ mov x0,op0;  
        ldr x1,m1;  
        cmp x0,x1;  
        cset x0,lt ],  
    x0)
```


Compiling an expression (function call)

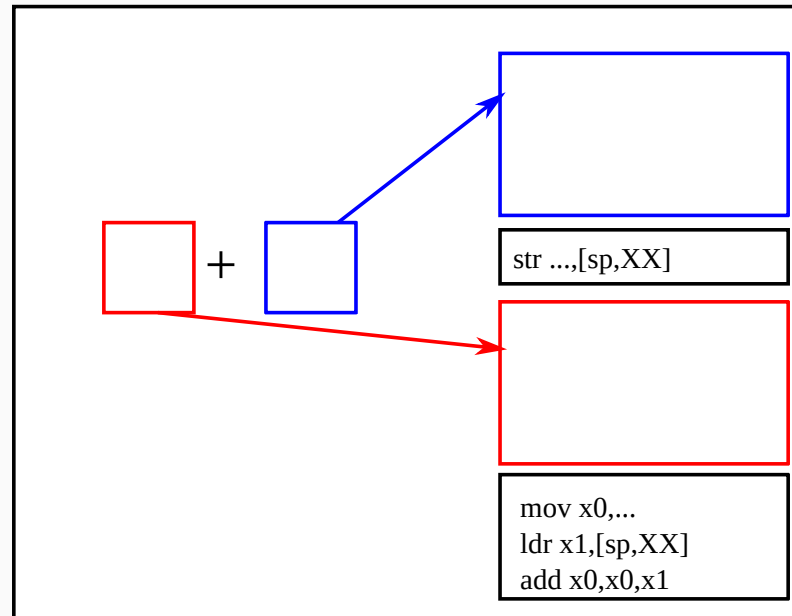
- $\approx$ instructions for all arguments; put them to positions specified by ABI; a `bl` instruction



```
gen_call_expr (ExprCall(f, [e0;e1;...])) ... =  
  [(i0,op0);(i1,op1);...],[m0;m1;...]  
  = gen_exprs [e0;e1;...] ...;  
  ( (i0 + [str op0,m0])  
    + (i1 + [str op1,m1])  
    + ...  
    + [ldr x0,m0;  
      ldr x1,m1;  
      ...;  
      bl f],  
    x0)
```

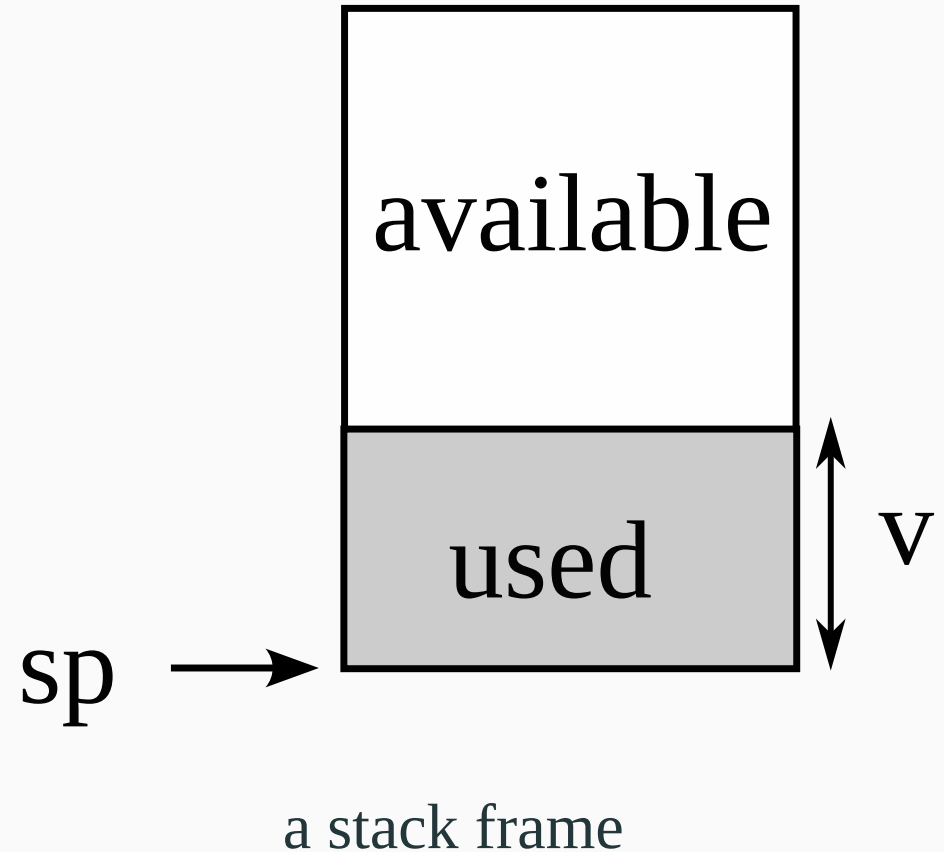
A few left-out details

- how to determine locations to save values of *subexpressions* and *variables*?
- that is, how to determine XX below:



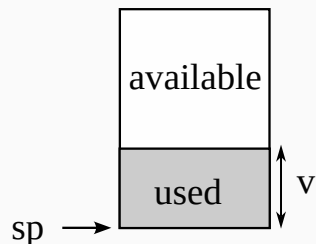
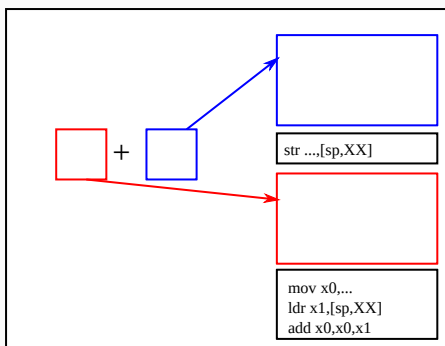
Determining where to save subexpressions

- “gen_expr E ” receives a value (v) pointing to the lowest end of free space in the current stack frame
- “gen_expr $E \ v \ \dots$ ” generates instructions that evaluate E using (destroying) only addresses at or above $SP+v$



Determining where to save subexpressions

- when evaluating $A + B$,
 1. evaluate B , using $SP+v$ and higher; save the result at $SP+v$
 2. evaluate A , using $v + 8$ and higher addresses



```
gen_add_expr ExprOp("+",[e0;e1]) v ... =
  insns1,op1 = gen_expr e1 v ... ;
  insns0,op0 = gen_expr e0 (v+8) ... ;
  ( insns1
    + [ str op1,[sp,v] ]
    + insns0
    + [ mov x0,op0;
        ldr x1,[sp,v];
        add x0,x0,x1],
    x0)
```

Locations to hold variables

```
if (...) {  
  long a, b, c;  
  ...  
}
```

we obviously need to store a , b , c somewhere, but where?

- the problem is almost identical to saving values of subexpressions
- $\rightarrow$ `gen_stmt` also takes v pointing to the free space
- `(gen_stmt  $S$   $v$  ...)` generates instructions to execute S , using only addresses at or above $SP + v$
- $\Rightarrow$
 - $a \mapsto (SP + v)$
 - $b \mapsto (SP + v + 8)$
 - $c \mapsto (SP + v + 16)$

Environment: records where variables are held

- variable locations must be known when generating code for expressions referencing them
 - e.g., to compile $x + 1$, we need to know where x is stored
- $\Rightarrow$ make a data structure that holds a mapping: *variable* $\mapsto$ *location* (*environment*) and pass it to `gen_stmt` and `gen_expr`
 - generating code for variables look up the environment
 - a compound statement (`{ ... }`) adds new mappings to the environment

gen_expr receives an environment

```
gen_expr (ExprId(x)) env v =  
  m = env_lookup x env;  
  ([ ldr x0,m ], x0)
```

`env_lookup x env` searches environment `env` for `x` and returns its location

gen_stmt receives an environment too

```
gen_stmt (StmtCompound(decls, stmts)) env v =  
  env', v' = env_extend decls env v;  
  gen_stmts stmts env' v' ...
```

- `env_extend decls env v`:
 - assigns locations ($v, v + 8, v + 16, \dots$) to variables declared in `decls`
 - registers them in `env`
 - returns the new environment `env'` and the new free space `v'`

Implementing environment

- an environment is a list of (*variable name*, *location*) pairs (*association list*)
- `v = env_lookup x env`
 - returns the location paired with `x` in environment `env`
- `env' = env_add x v env`
 - returns a new environment `env'` which has a new mapping $x \mapsto v$ in addition to `env`
- `(env', v') = env_extend decls v env`
 - can be easily built on `env_add` (left for you)

Intermediate Representation (IR)

Intermediate Representation (IR)

- a common representation of programs used by a compiler
- roughly $\approx$ an assembly with unlimited variables
- purposes
 1. achieve portability
 - hopefully independent from the source language (C, C++, Rust, Go, Julia, etc.)
 - hopefully independent from the target language (x86, ARM, PowerPC, etc.)
 2. formulate optimizations as IR $\rightarrow$ IR transformations
- **note:** in the exercise you could design your IR, but it is not necessary (it is possible to directly go from AST $\rightarrow$ asm)

Optimizations performed on IR level (a brief)

- **constant folding and propagation** — compute values at compile time where possible
- **hoisting** — lift instructions in a loop outside of it
- **function call inlining** — replace a call to a function with its body
- **register allocation** — assign registers to variables to reduce memory access